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Trader Behavior vs Market Sentiment

An Analysis of Fear and Greed in Bitcoin Trading

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Tools Used: Python, Pandas, Matplotlib, Google Colab
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1. summary

1.1 Background

Financial markets are heavily influenced by collective psychology. In cryptocurrency markets, where volatility is high, trader behavior often reflects broader emotional states such as fear and greed.

1.2 Objective of the Study

The objective of this analysis is to examine whether and how trader behavior on Hyperliquid changes under different market sentiment regimes—specifically Fear and Greed—and to identify patterns that can inform better trading and risk management decisions.

1 Introduction

Purpose

In this project, I analyzed how trader behavior on Hyperliquid changes under different market sentiment conditions, specifically **Fear** and **Greed**.

By combining historical trading activity with Bitcoin market sentiment data, I aimed to understand how factors such as profitability, trading volume, and execution behavior vary across sentiment regimes and what this means for real trading decisions.

2 Datasets Used

Bitcoin Fear & Greed Index

This dataset was used to capture overall market psychology on a daily basis.

To keep the analysis aligned with the objective, the original sentiment levels were mapped into two categories: **Fear** and **Greed**, while neutral days were excluded.

Hyperliquid Historical Trader Data

This dataset contains executed trades from Hyperliquid, including timestamps, trade sizes, and realized PnL.

Trade-level data was aggregated into daily metrics to make it directly comparable with daily sentiment values.

3 Methodology

I first converted individual trade records into daily-level summaries in order to reduce noise and focus on behavioral trends.

The following metrics were computed for each day:

- Total realized PnL
- Total trading volume
- Win rate (percentage of profitable trades)
- Number of trades executed

The daily trader metrics were then merged with the sentiment dataset using the date as the common key.

Finally, I compared trader behavior across Fear and Greed periods using summary statistics and visual distributions.

4️⃣ Key Trading Insights

🔹 Insight 1: Trading Activity Increases During Greed, but Efficiency Does Not

What I observed:

Greed days consistently show higher trading volume and a larger number of trades compared to Fear days.

What this suggests:

This behavior likely reflects increased optimism and FOMO-driven participation, where traders trade more frequently rather than more selectively.

Why it matters for trading:

Higher activity alone does not guarantee better outcomes. During Greed phases, maintaining discipline in position sizing and trade selection becomes especially important.

◇ Insight 2: Fear Periods Are Associated with Lower Win Rates and Sharper Losses

What I observed:

Win rates tend to drop during Fear periods, and the PnL distribution shows more extreme negative outliers.

What this suggests:

This pattern points toward panic-driven exits and reduced patience, where traders are more likely to close positions under stress.

Why it matters for trading:

During Fear regimes, conservative risk management—such as smaller position sizes and stricter exit rules—can help limit drawdowns.

◇ Insight 3: Profitability Becomes More Volatile During Greed

What I observed:

PnL distributions during Greed periods display wider variance, with both large gains and significant losses.

What this suggests:

Traders appear to take on higher risk during optimistic market conditions, leading to inconsistent performance across days.

Why it matters for trading:

Greed phases tend to reward fewer, high-conviction trades rather than frequent execution with elevated risk.

◇ Insight 4: Win Rate Alone Does Not Explain Performance

What I observed:

Even when win rates remain moderate, overall daily PnL can differ substantially between Fear and Greed regimes.

What this suggests:

Profitability is influenced more by trade sizing and timing than by the percentage of winning trades alone.

Why it matters for trading:

Traders should focus on risk–reward balance and expected value instead of relying solely on win rate as a performance metric.

◇ Insight 5: Market Sentiment Functions as a Risk Regime Indicator

What I observed:

Clear shifts in trading behavior occur when sentiment moves between Fear and Greed.

What this suggests:

Rather than acting as a direct price signal, sentiment serves as a macro-level indicator of market risk conditions.

Why it matters for trading:

Adapting strategy parameters based on sentiment regime can lead to more consistent performance across different market environments.

6 Conclusion

From this analysis, it is clear that trader behavior on Hyperliquid is strongly influenced by overall market sentiment.

Greed periods tend to amplify trading activity and risk-taking, while Fear periods expose weaknesses in execution and risk control.

Incorporating sentiment-aware adjustments into trading strategies can help traders navigate these behavioral shifts more effectively.

7 Limitations & Future Work

One limitation of this analysis is the absence of leverage data, which restricted deeper risk-based evaluation.

Future extensions of this work could include:

- Segmenting traders based on behavior and performance
- Analyzing strategy performance during transitions between Fear and Greed
- Incorporating risk-adjusted performance metrics such as Sharpe ratio and drawdowns